

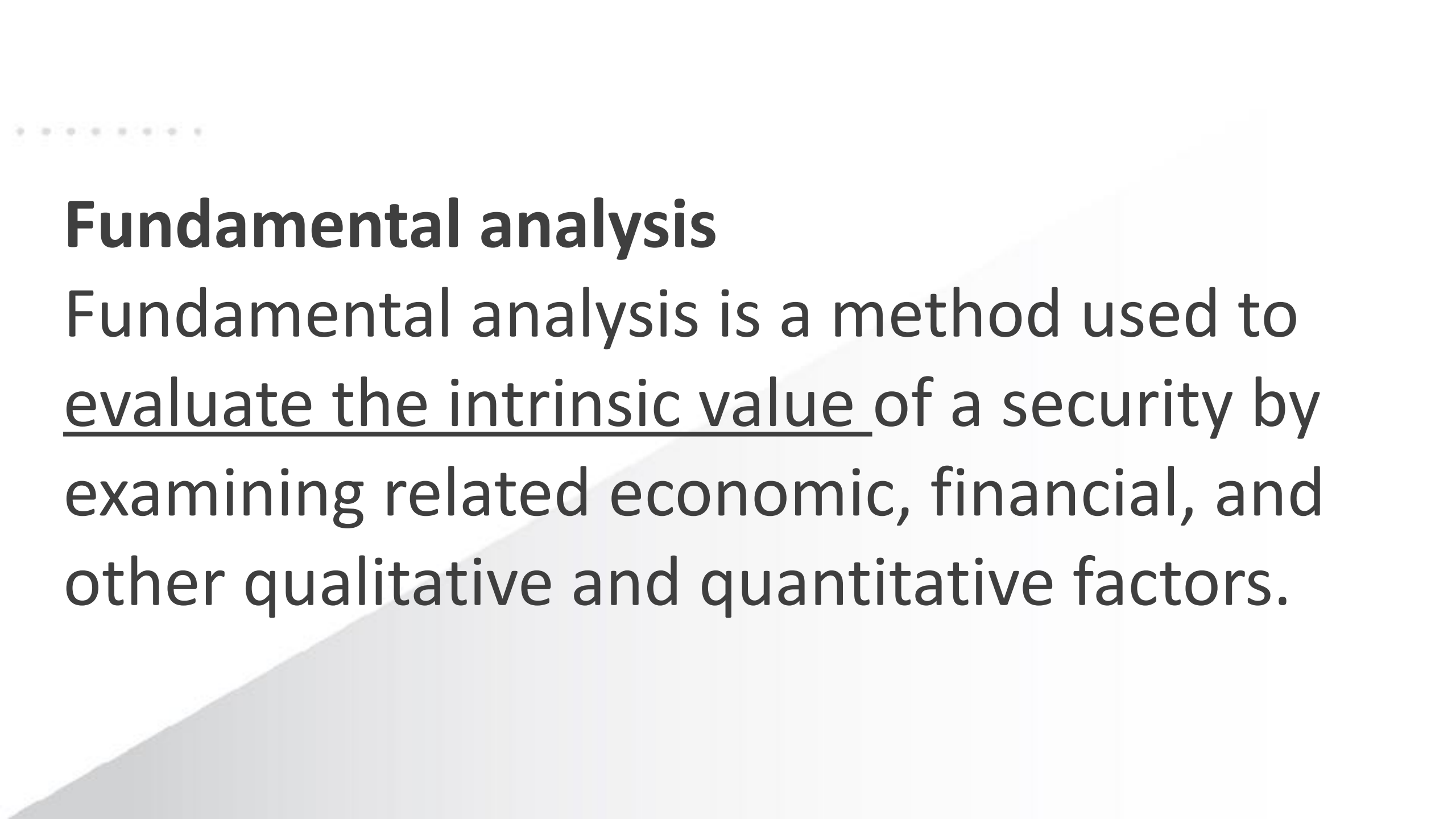


The origin of Portfolio risk management

A series of ten small, light gray dots arranged in a horizontal line at the top left of the slide.

“The central question for investment management practitioners is: Where should I invest my money?”

— Sebastien Lleo

A decorative graphic in the top-left corner consisting of a horizontal row of eight dots, with the first seven being light gray and the eighth being white. Below the dots is a large, light gray triangle pointing towards the bottom-right corner of the slide.

Fundamental analysis

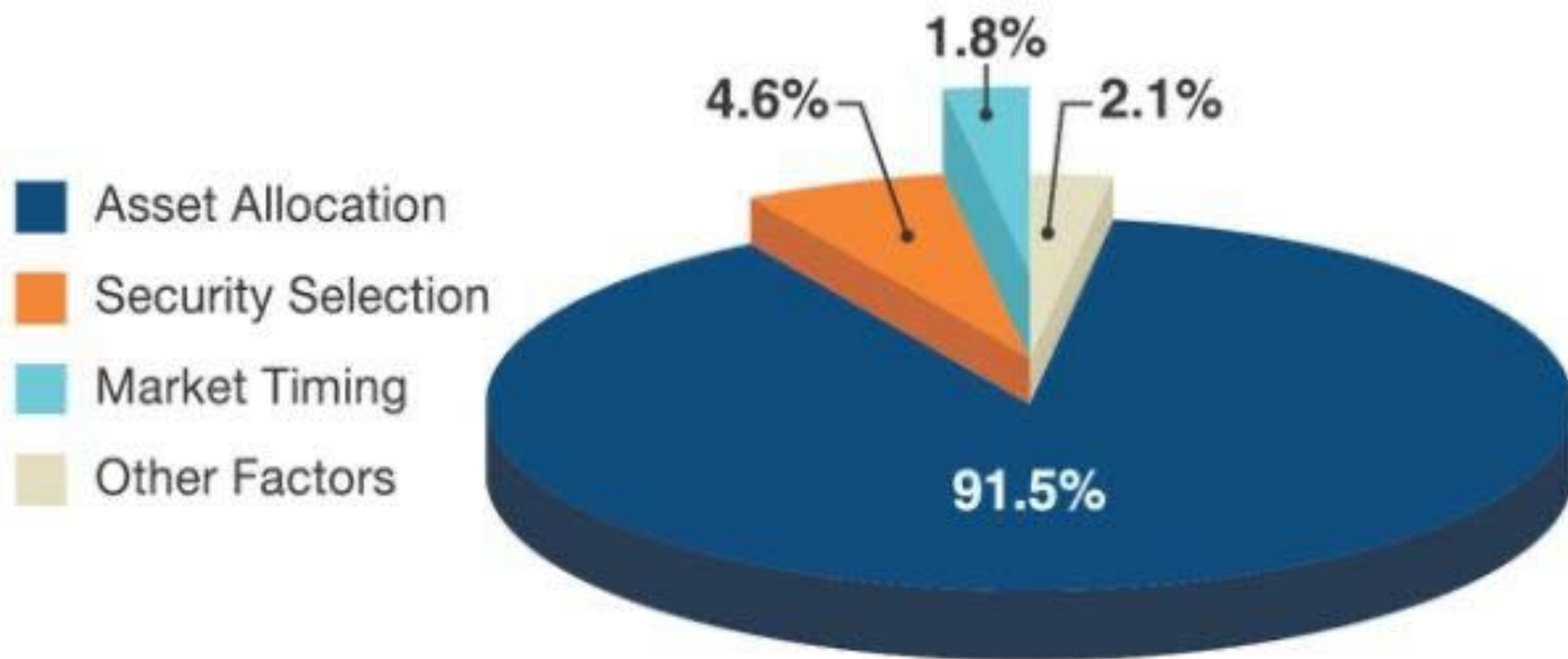
Fundamental analysis is a method used to evaluate the intrinsic value of a security by examining related economic, financial, and other qualitative and quantitative factors.



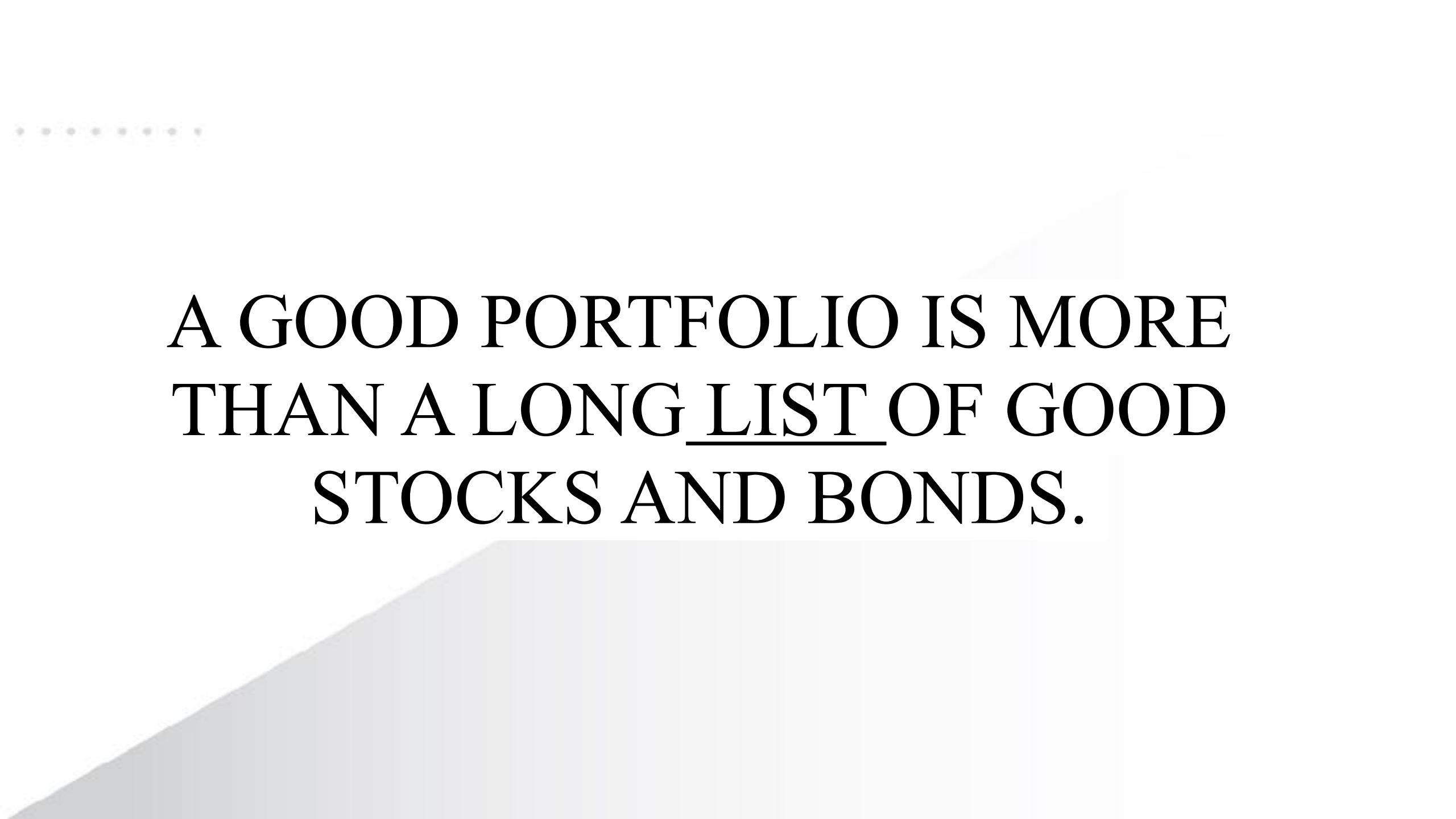
Technical analysis?

Technical analysis is a method used to evaluate and forecast the price movements of securities, such as stocks, by analyzing historical price and volume data.

Drivers of investment results



Source: Financial Analyst Journal, 1986

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A GOOD PORTFOLIO IS MORE
THAN A LONG LIST OF GOOD
STOCKS AND BONDS.

RISK AND RETURN



"I wanna maximize return"



"The hypothesis that the investor should maximize discounted return must be rejected." - Markowitz (1952)

.....

ผลตอบแทนที่คาดหวัง

● A

● C

● B

ความเสี่ยง

เราควรเลือกลงทุนที่จุด A B หรือ C

Portfolio Selection is a Two-Stage



The process of selecting a portfolio may be divided into two stages.

- The first stage starts with observation and experience and ends with beliefs about the future performances of available securities.
- The second stage starts with the relevant beliefs about future performances and ends with the choice of portfolio.

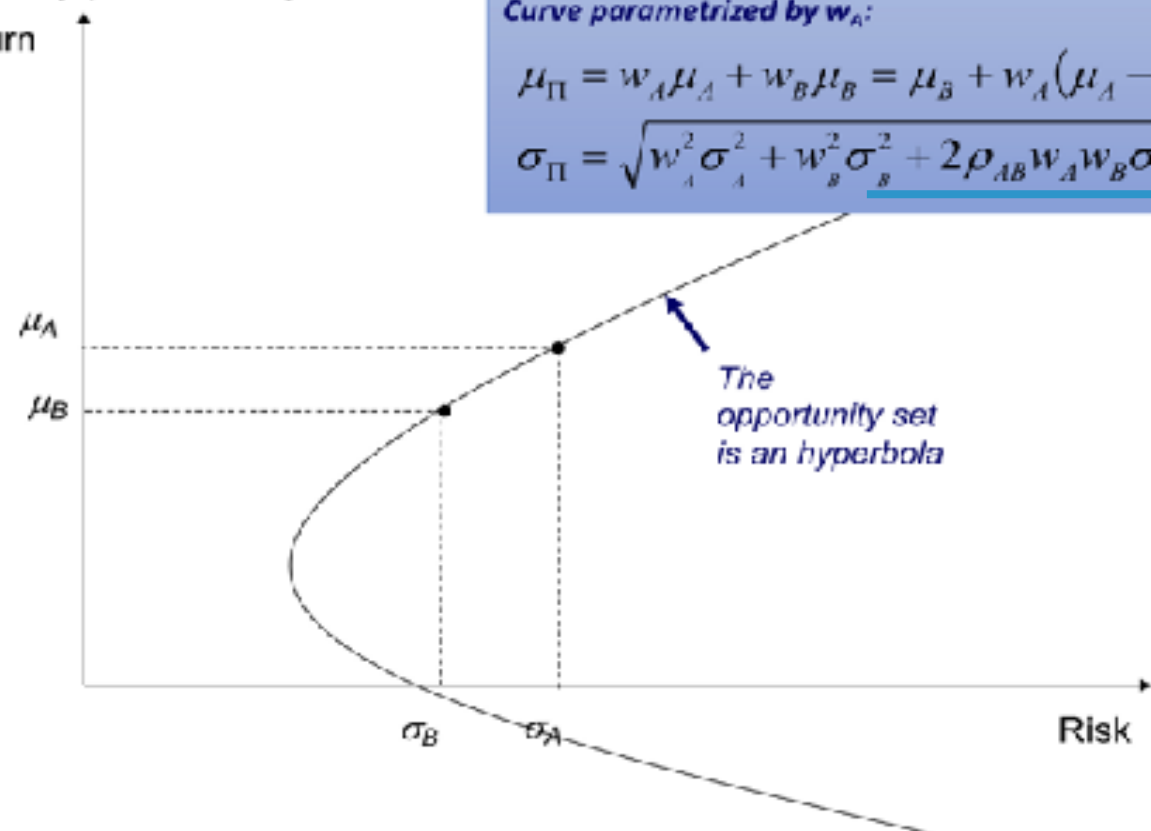


Historical note

Although “**Don’t put all your eggs in the same basket**” was a popular say in the investment world even before Markowitz, portfolios tended to be constructed as collections of individual securities selected for their return potential and with little regards for their risks or interactions.

The opportunity set

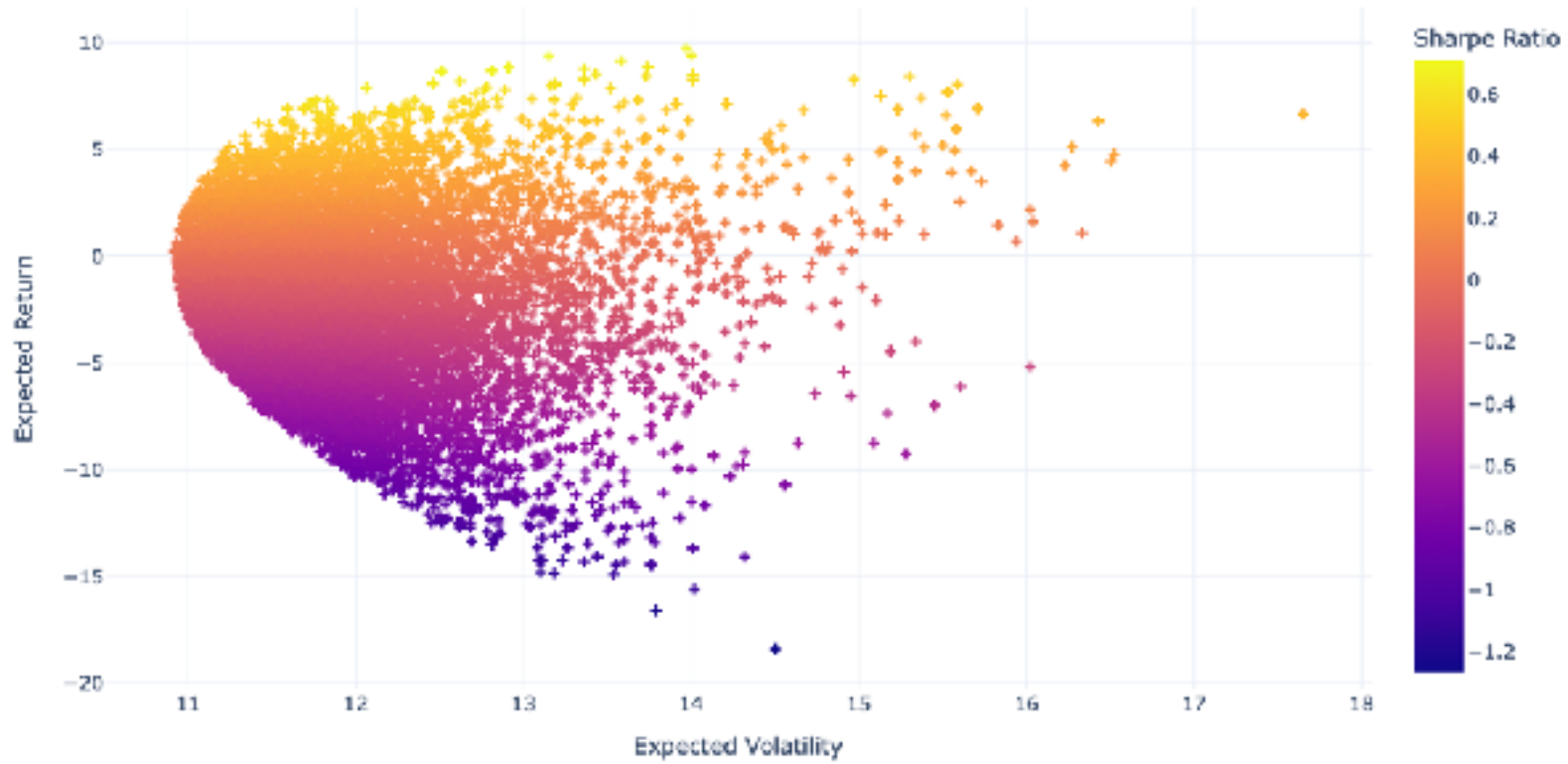
Return



Monte Carlo Portfolio

Random Portfolio

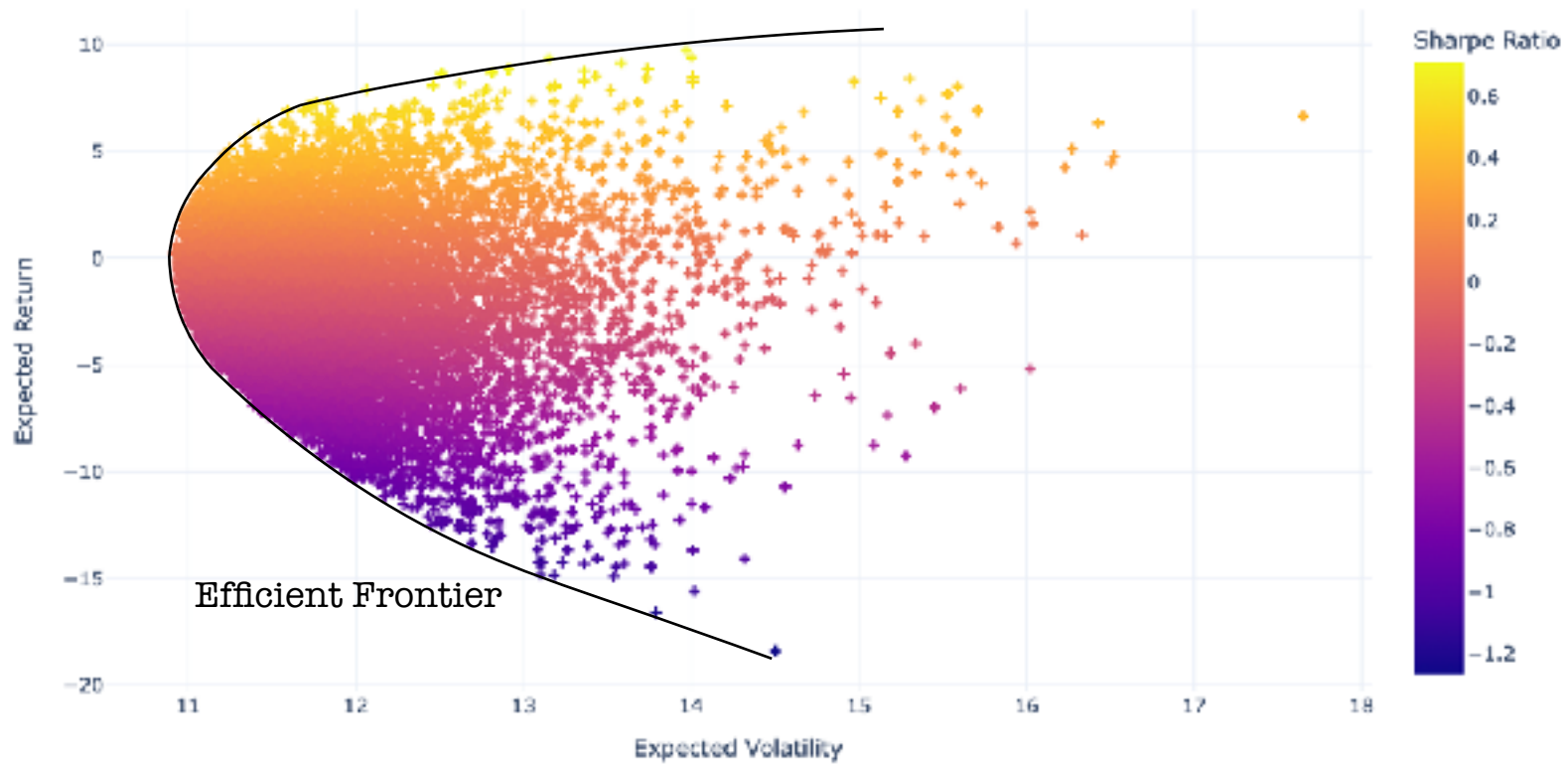
Monte Carlo Portfolio Simulation



* 10,000 Random Portfolio

Random Portfolio

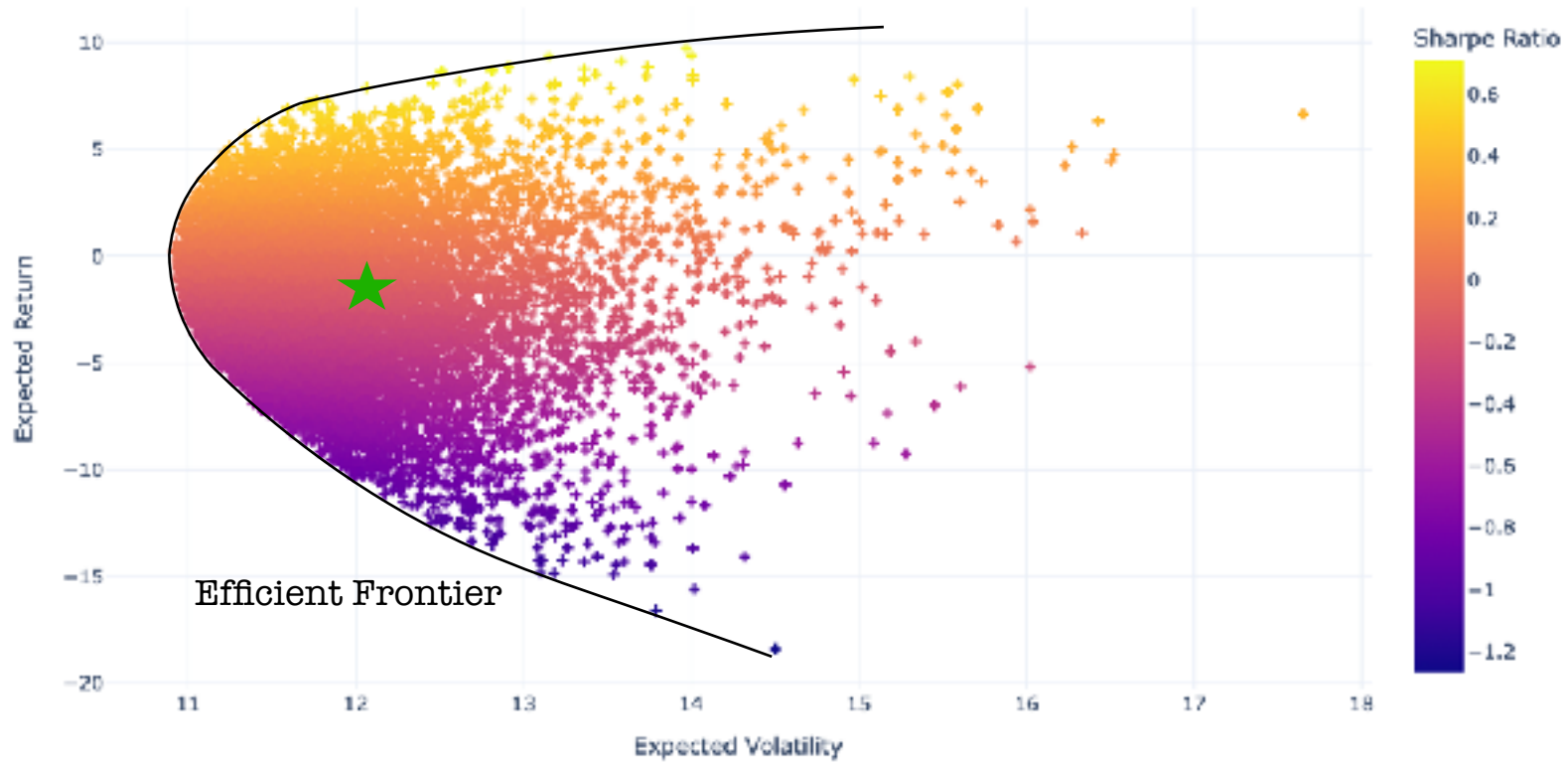
Monte Carlo Portfolio Simulation



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Random Portfolio

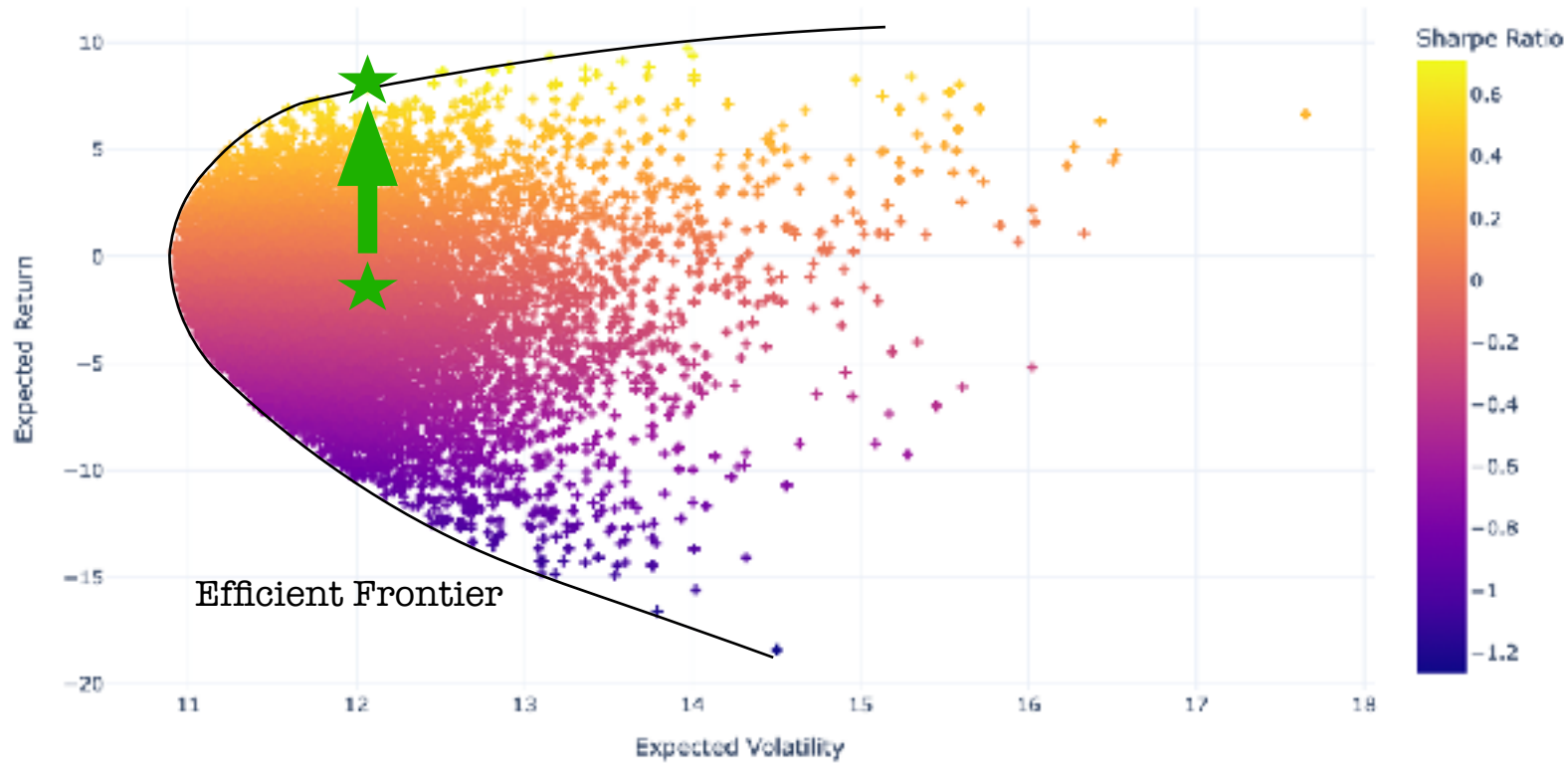
Monte Carlo Portfolio Simulation



* 10,000 Random Portfolio

Random Portfolio

Monte Carlo Portfolio Simulation

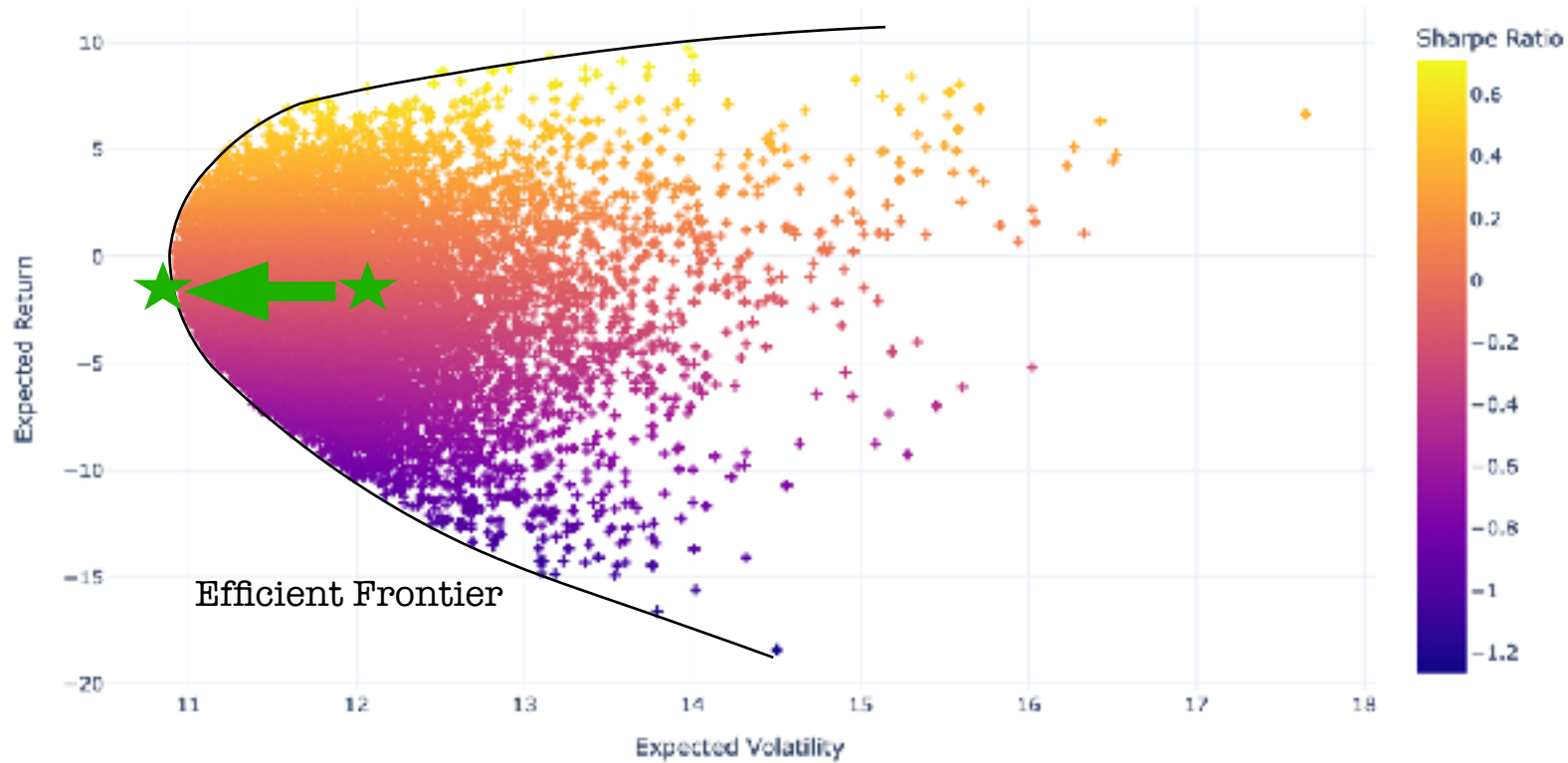


Increase Return

* 10,000 Random Portfolio

Random Portfolio

Monte Carlo Portfolio Simulation

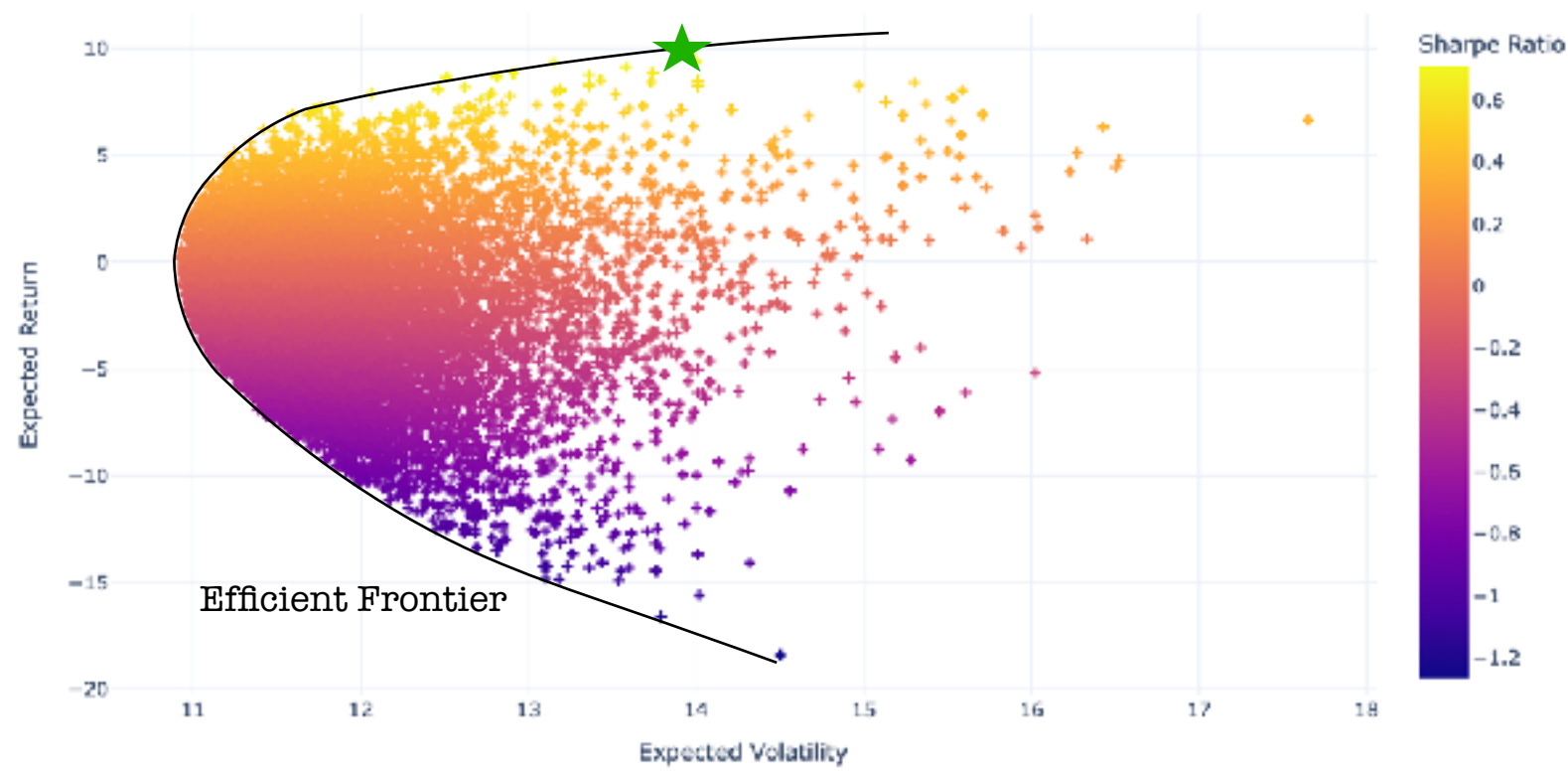


Decrease Volatility

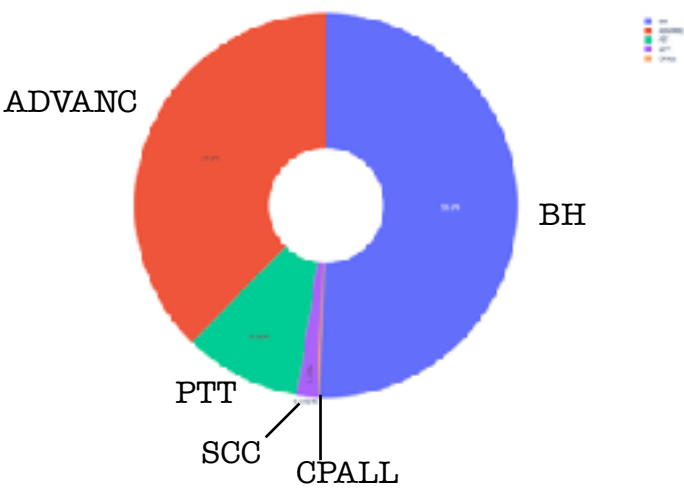
* 10,000 Random Portfolio

Random Portfolio

Monte Carlo Portfolio Simulation



Max Return Portfolio

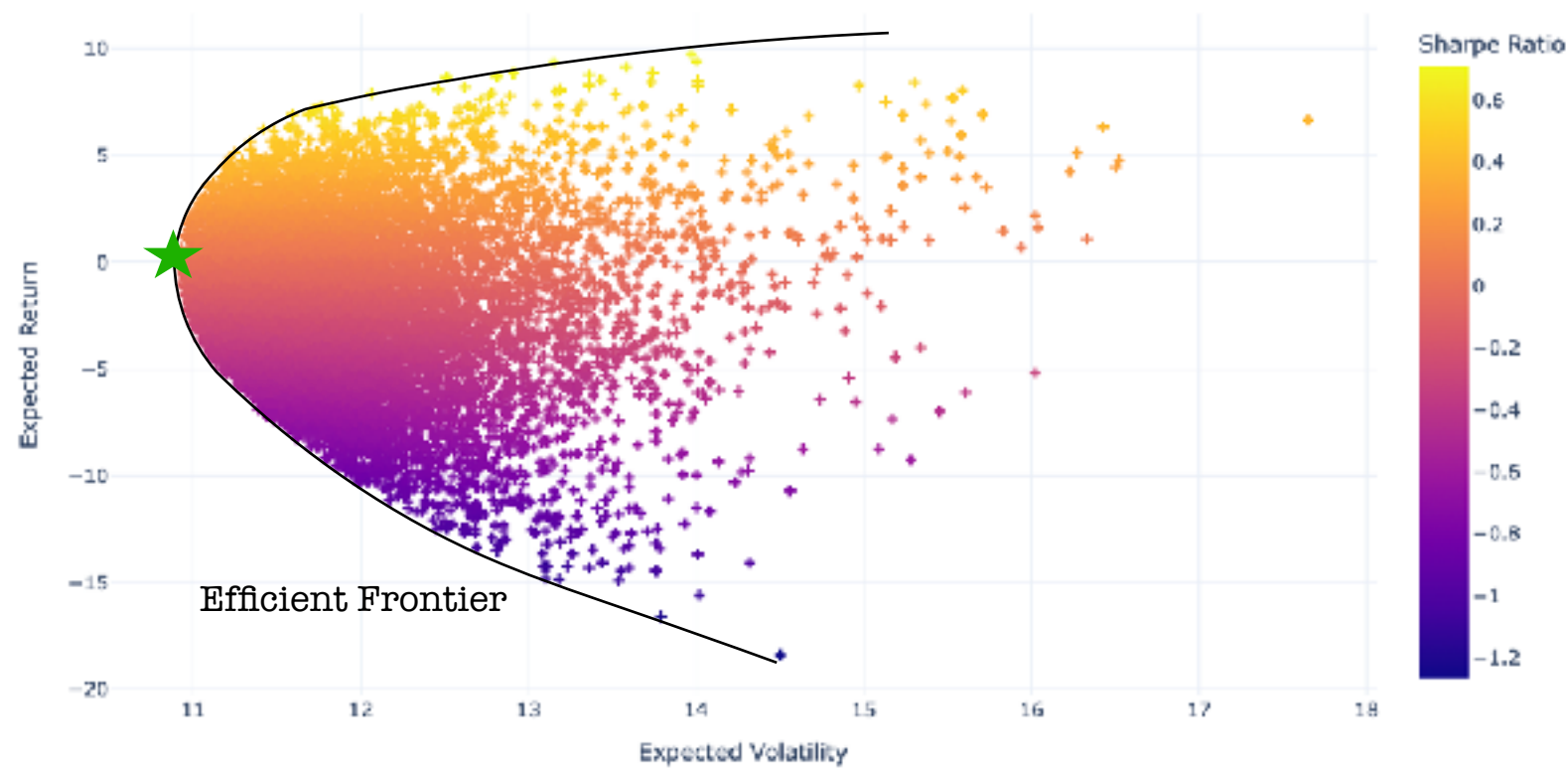


Return	9.73%
Volatility	13.97%
Sharpe Ratio	0.7

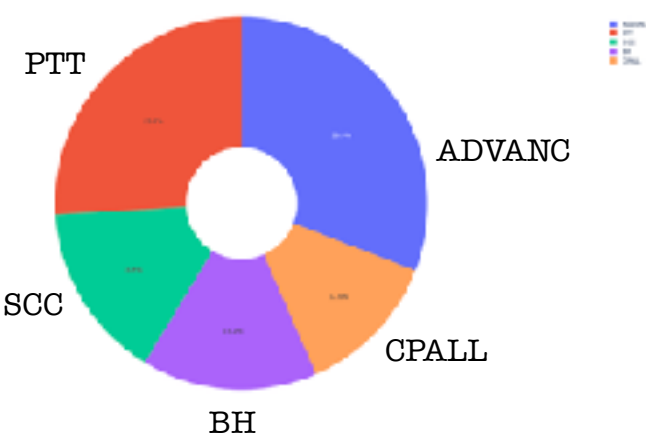
* 10,000 Random Portfolio

Random Portfolio

Monte Carlo Portfolio Simulation



Min Volatility Portfolio

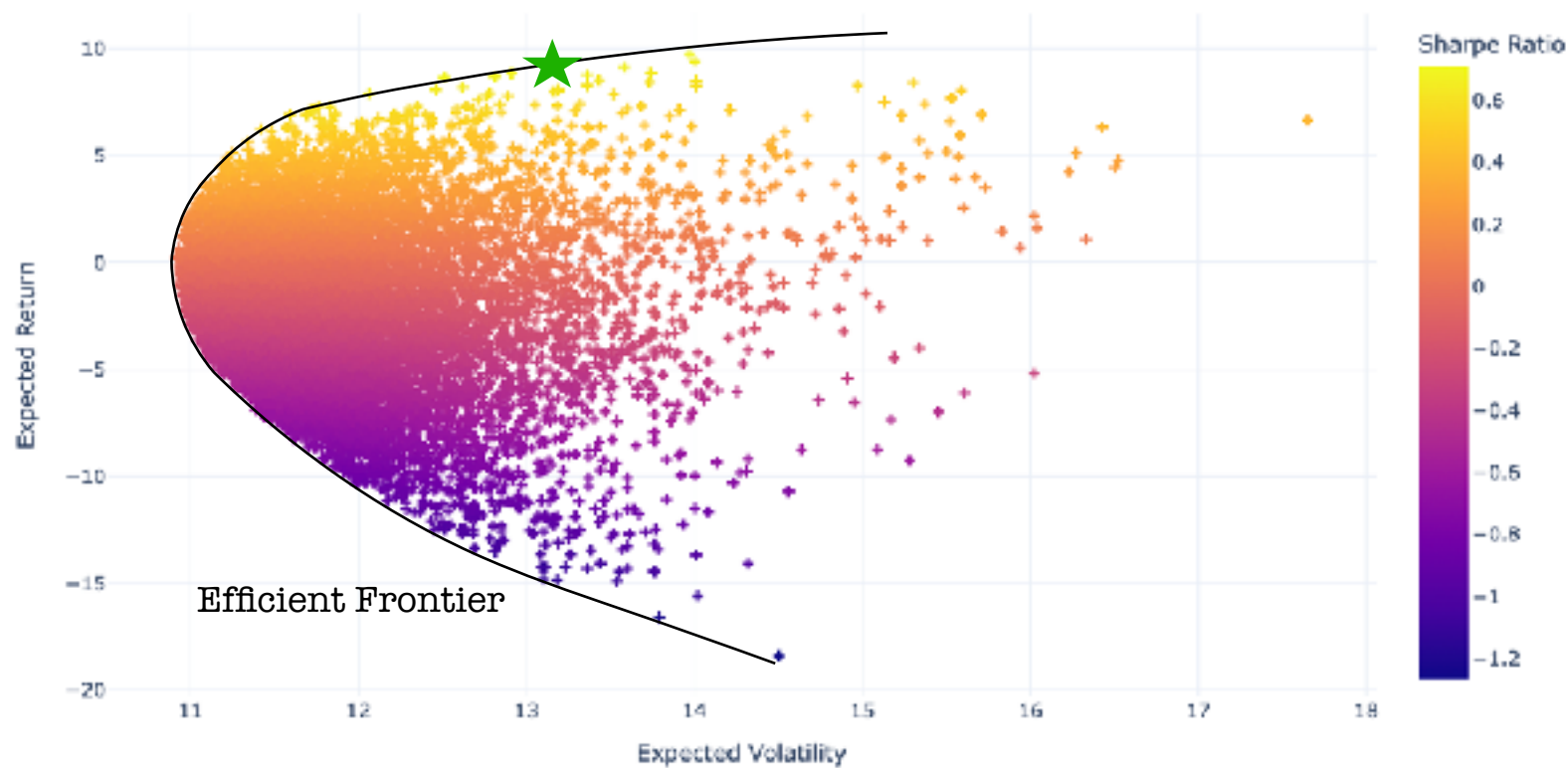


Return	0.22%
Volatility	10.92%
Sharpe Ratio	0.02

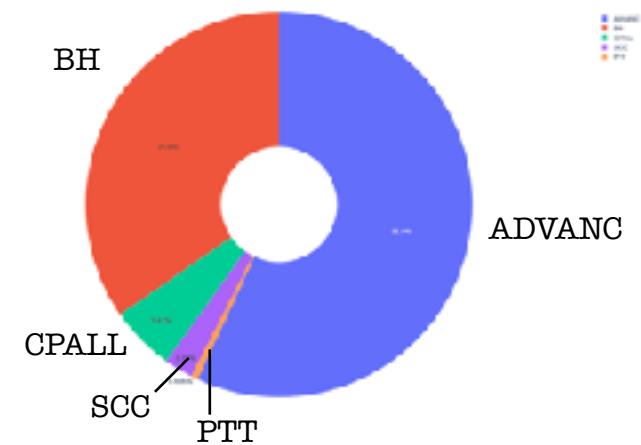
* 10,000 Random Portfolio

Random Portfolio

Monte Carlo Portfolio Simulation



Max Sharpe Ratio Portfolio



Return	9.35%
Volatility	13.15%
Sharpe Ratio	0.71

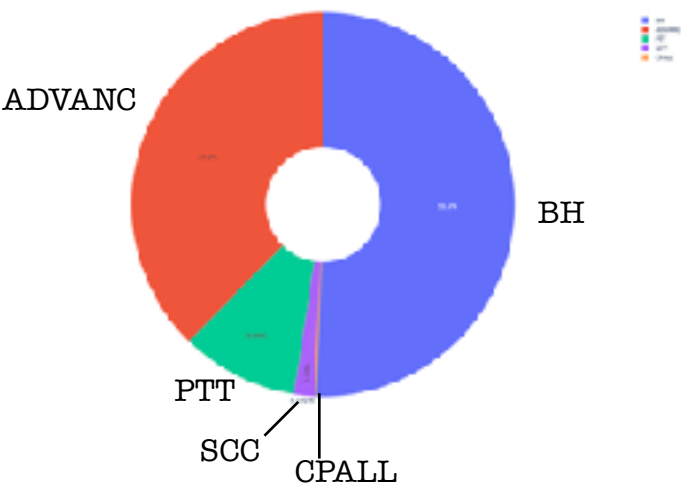
* Max Risk-Adjusted Return Portfolio

* Sharpe Ratio = $\frac{\text{Annualized Return}}{\text{Annualized Volatility}}$

* 10,000 Random Portfolio

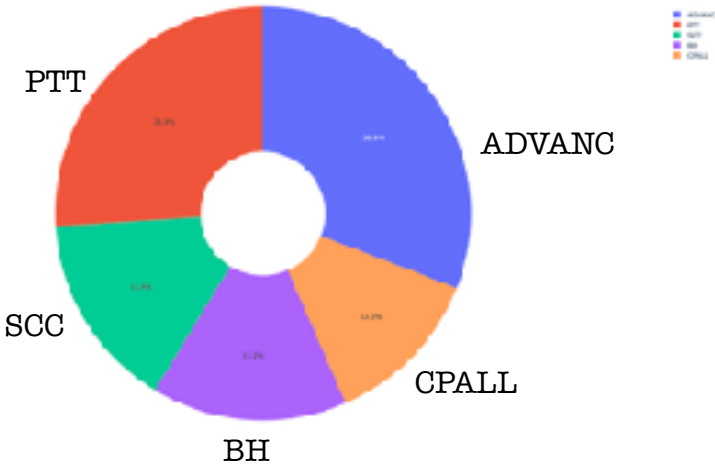
Random Portfolio

Max Return Portfolio



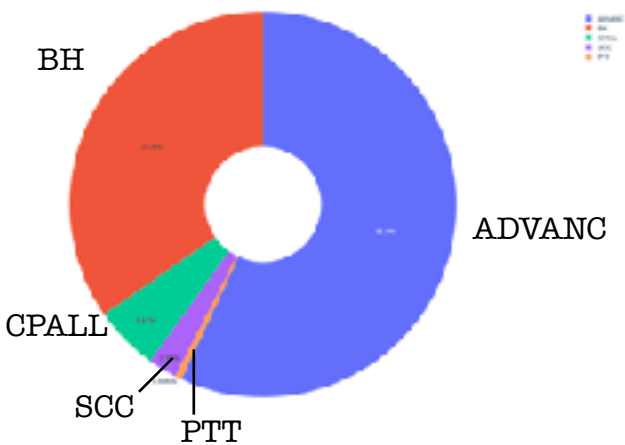
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Min Volatility Portfolio



Return	0.22%
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Max Sharpe Ratio Portfolio



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* 10,000 Random Portfolio

Mean-Variance Portfolio

Objectives

Maximize Return in Portfolio

$$\max_w w^T \mu$$

Minimize Volatility in Portfolio

$$\min_w w^T \Sigma w$$

Minimize Volatility and Maximize Return
in Portfolio

$$\min_w w^T \Sigma w - w^T \mu$$

Minimize Volatility and Maximize Return
with Risk Aversion in Portfolio

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma w^T \mu$$

μ : Return in each stock

Σ : Covariance matrix

γ : Risk aversion parameters

w : Weights

Covariance Matrix [n x n]

 Σ

Check

Positive Definite

All eigen value (λ) > 0

Positive Semi-Definite

All eigen value (λ) ≥ 0

$$* A \vec{u} = \lambda \vec{u}$$

Covariance Matrix [n x n]

 Σ

Check

Positive Definite

All eigen value (λ) > 0

Positive Semi-Definite

All eigen value (λ) ≥ 0

$$* A \vec{u} = \lambda \vec{u}$$

If NOT !!

Upgrade Covariance Matrix

Find Nearest Positive - Semi Definite

Covariance Matrix [n x n]

 Σ

Check

Positive Definite

All eigen value (λ) > 0

Positive Semi-Definite

All eigen value (λ) ≥ 0

$$* A \vec{u} = \lambda \vec{u}$$

If YES !!

Noise Reduction

- > Fourier Transform
- > Shrinkage Technique
- > Random Matrix Theory

Constraint

Long-only Portfolio

$$w \geq 0$$

Total weights equal to total wealth

$$\sum_{i=1}^N w = 1$$

Target Return

$$w^T \mu = \text{constant}$$

Target Volatility

$$w^T \Sigma w = \text{constant}$$

μ : Return in each stock

Σ : Covariance matrix

w : Weights

Portfolio Optimization

Minimize Volatility and Maximize Return
in Portfolio

Objective Function :

$$\min_w w^T \Sigma w - w^T \mu$$

Long-only Portfolio

Total weights equal to total wealth

Constraint :

$$0 \leq w_i \leq 1$$

$$\sum_{i=1}^N w_i = 1$$

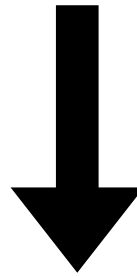
Lagrange Multiplier Optimization

Objective

+ λ

Constraint

$$\mathcal{L}(w, \lambda_0, \lambda) = w^T \Sigma w - w^T \mu + \lambda_0 (\mathbf{1}^T w - 1) + \lambda^T w$$



Partial Differentiation (Calculus)

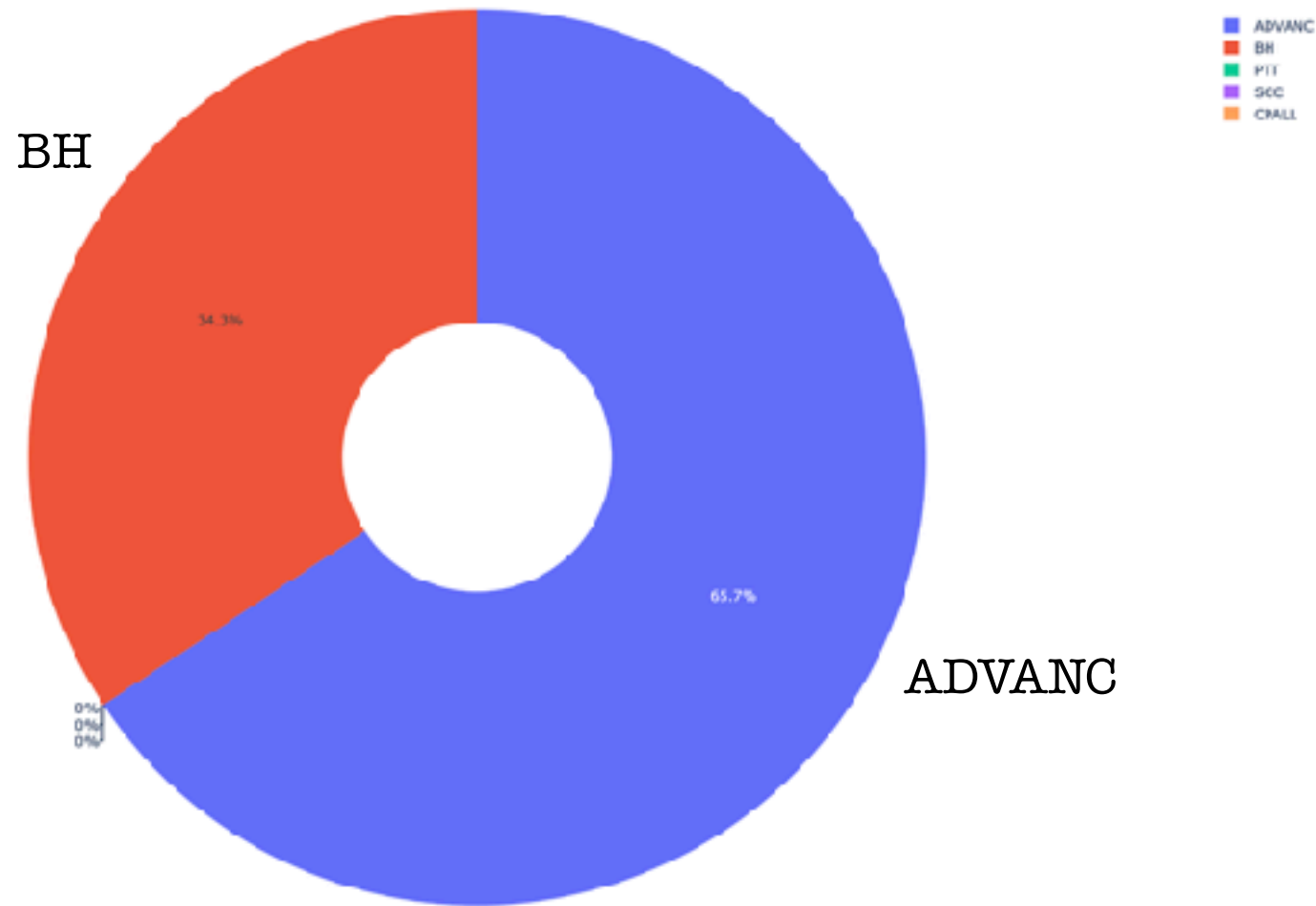
$$\frac{\partial \mathcal{L}(w, \lambda_0, \lambda)}{\partial w} = \frac{\partial \mathcal{L}(w, \lambda_0, \lambda)}{\partial \lambda_0} = \frac{\partial \mathcal{L}(w, \lambda_0, \lambda)}{\partial \lambda} = 0$$

Optimal Weights Solution

λ : Lagrange Multiplier

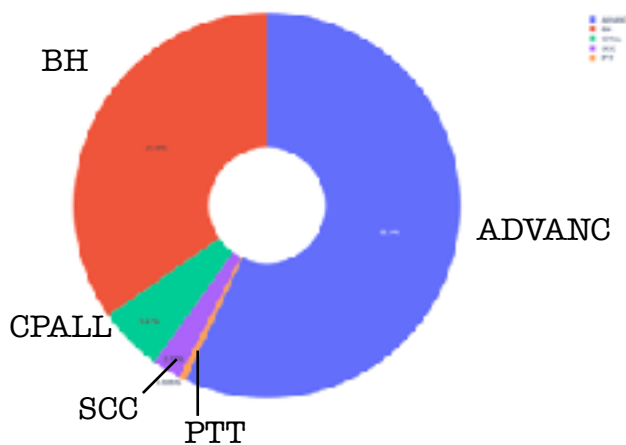
Max Sharpe Ratio Optimal Portfolio

Return	11.36%
Volatility	13.7%
Sharpe Ratio	0.83



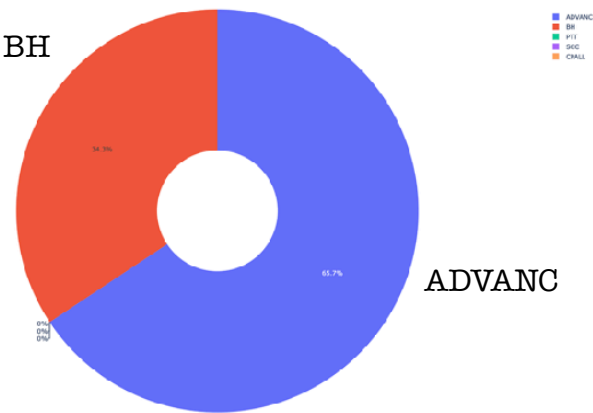
Max Sharpe Ratio Optimal Portfolio

Random 10,000 Portfolios



Return	9.35%
Volatility	13.15%
Sharpe Ratio	0.71

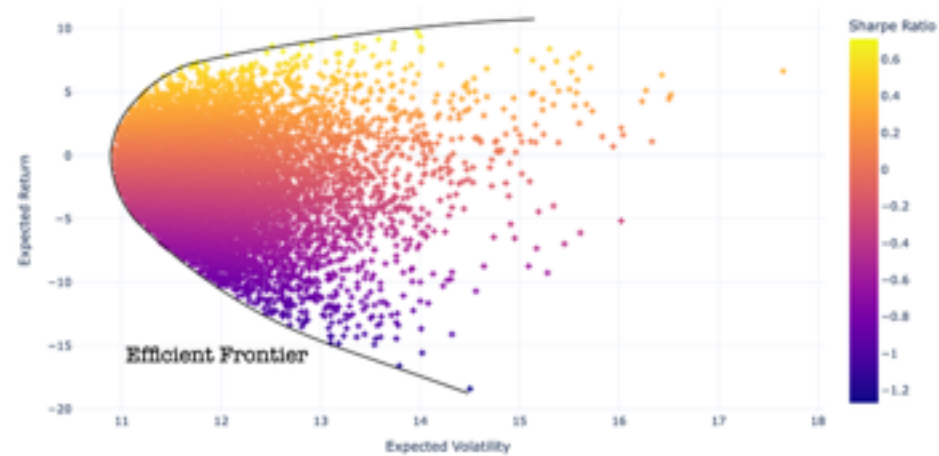
Mean-Variance Portfolios



Return	11.36%
Volatility	13.7%
Sharpe Ratio	0.83

Conclusion

Monte Carlo Portfolio Simulation



Pros

- > Know the logic of rational investors
- > Know the expected outcome
- > Can apply with many strategies

Cons

- > Unstable Weights Optimal
- > Depend on Asset's Characteristic
- > Can not deal with tail risk



All models are wrong,
but some are useful.

George E. P. Box

• However, historically MPT has suffered from two main drawbacks:

- Parameter estimation
- Dimensionality

A series of small, light gray dots are arranged in a horizontal line at the top left of the slide. A large, light gray, semi-transparent graphic, resembling a stylized mountain or a large arrow pointing upwards and to the right, occupies the lower half of the slide, serving as a background for the text.

“When **past performance** of securities is used as input, the output is a portfolio which performed particularly **well in the past.**”

— Harry Markowitz

A series of ten small, light gray dots arranged in a horizontal line at the top left of the slide.

“When the beliefs of security analysts are used as input, the output of the analysis is the implication of these beliefs”

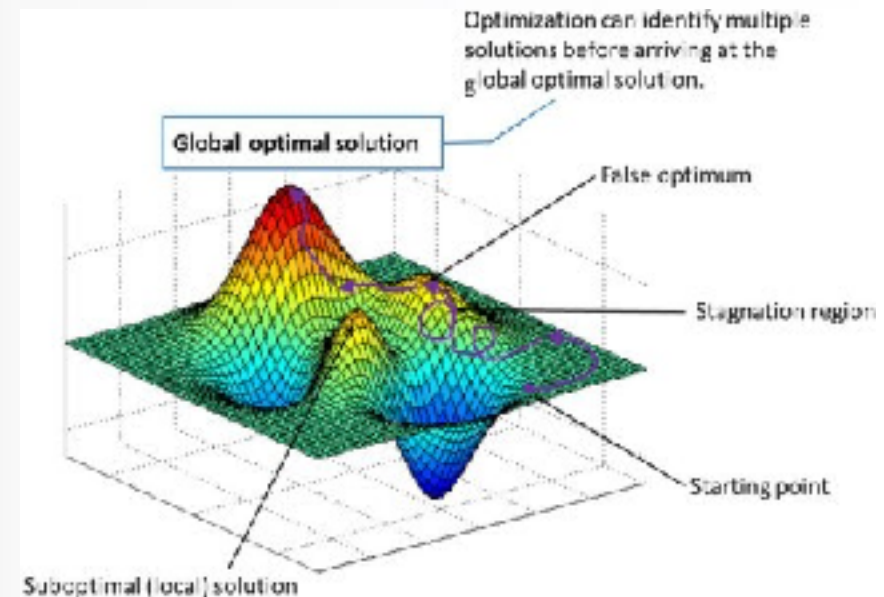
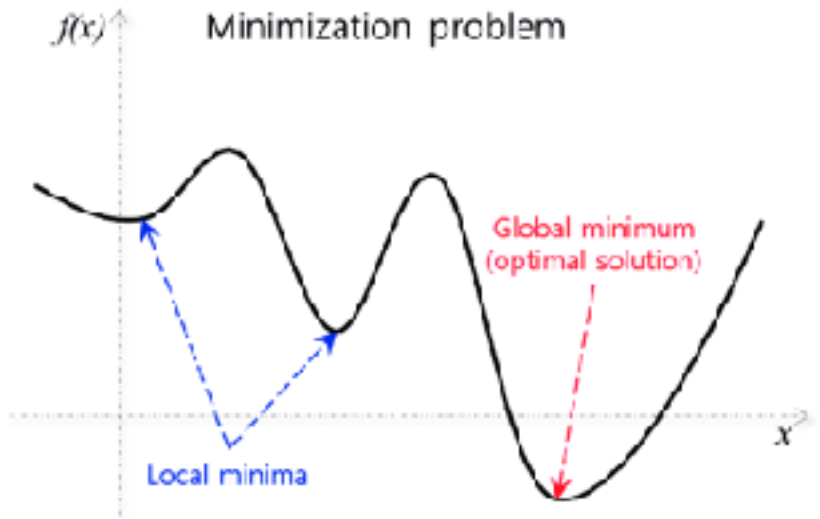
— Harry Markowitz

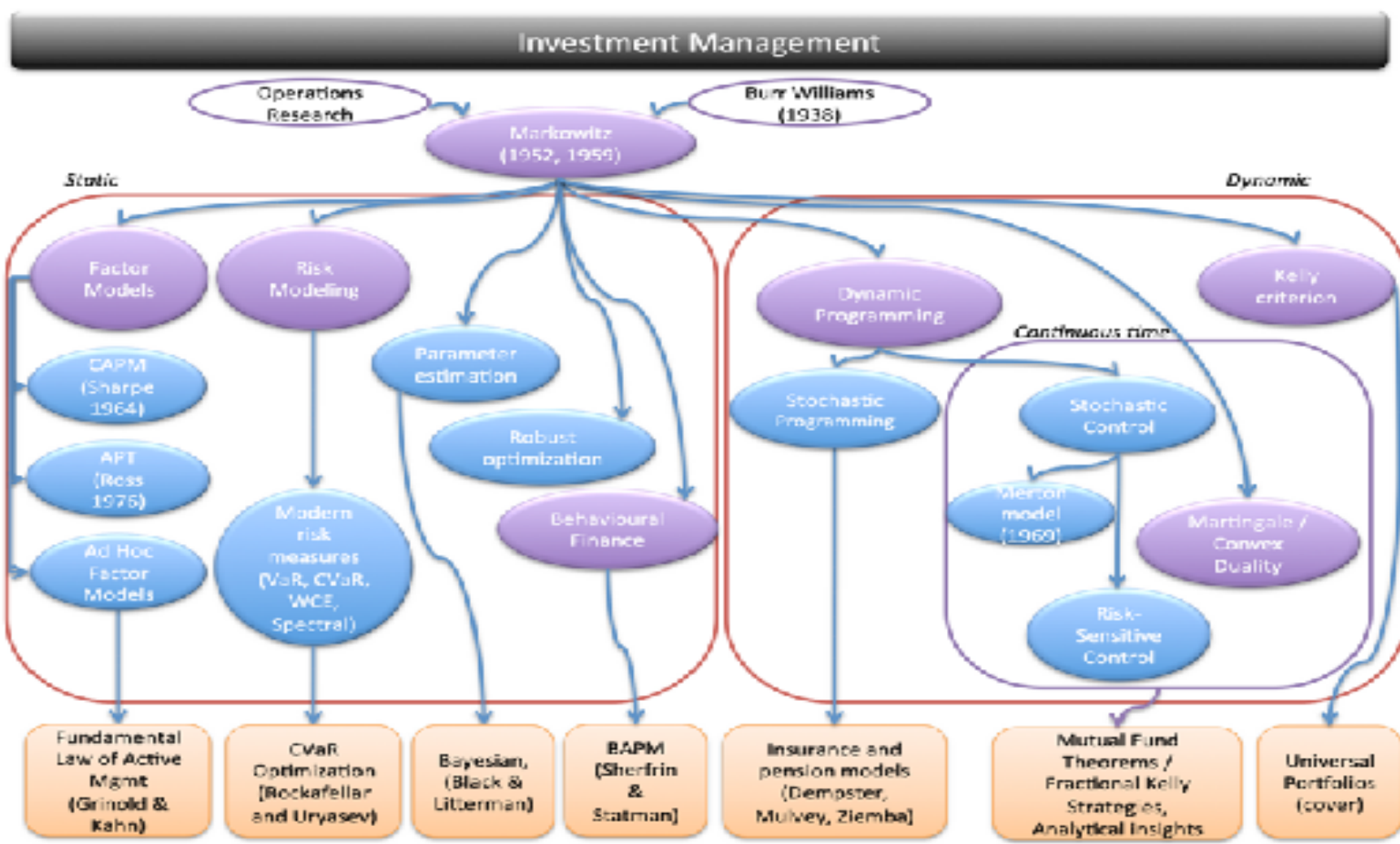


- Because optimizers are particularly efficient at taking advantage of the smallest discrepancy in data to reach their objective, **parameter estimation** is critical to get workable investment policies. This phenomenon is often called the “garbage in, garbage out” syndrome.
- The good news is that the variance and covariance of returns tend to be quite stable over long periods of time...
- ...but the bad news is that it would take hundreds of years of financial data to get a reasonably accurate estimates of expected returns.
 - few assets have been traded long enough;
 - in any case, market conditions change over time which cause “breaks” in the time series of returns.

Dimensionality

- **Dimensionality** was and is still an important concern due to the vast size of financial markets.
- Although factor models can be used to reduce the dimensionality of the problem, they still do not hold all the answers:





- 
- Other Risk Measure
 - Black-Littermann Portfolio
 - Dynamic Portfolio Optimization
 - Roy Safety first Asset Allocation
 - Hierarchical Risk Parity
 - Risk Parity Portfolio
 - Adaptive Allocation
 - Goal Based Portfolio
 - RL Based Portfolio

Risk-Budget Portfolio Optimization

Risk Decomposition :

$$\sigma(w) = \sum_{i=1}^N \frac{\partial \sigma(w)}{\partial w_i} w_i$$

$\sigma(w)$: Risk Measure

Value At Risk

Volatility

Risk Contribution :

$$\frac{\partial \sigma(w)}{\partial w_i} w_i$$

Risk-Budget

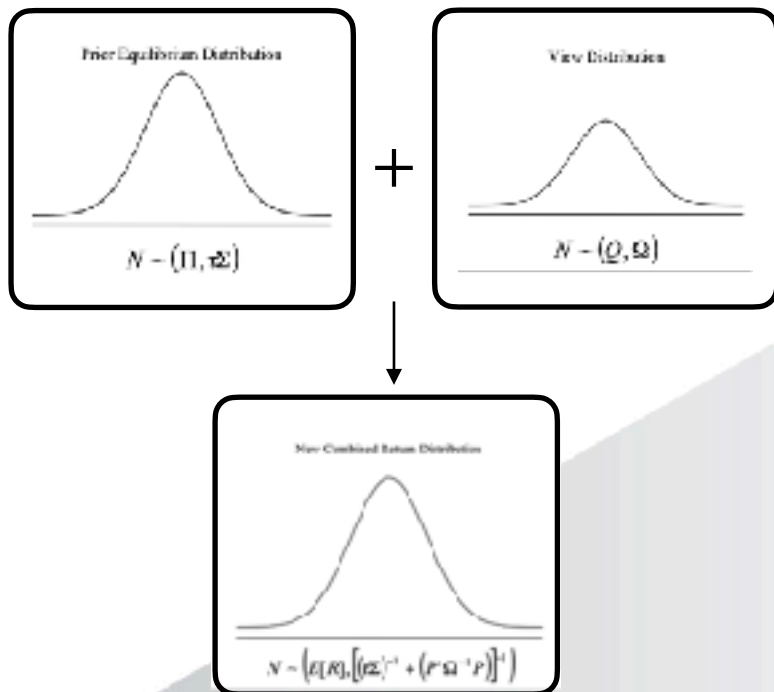
Objective Function :

$$\min_w \sum_{i=1}^N \sum_{j=1}^N \left(\frac{\partial \sigma(w)}{\partial w_i} w_i - \frac{\partial \sigma(w)}{\partial w_j} w_j \right)^2$$

Black-Littermann Portfolio Optimization

Objective Function :

$$E[R] = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1} [(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q]$$



Ω : Diagonal Matrix for View's Error

Π : Equilibrium Return

Q : Investor Views

P : Views Include

τ : Scalar



$$\Pi = \lambda \Sigma w_{eq}$$

λ : Risk Aversion

w_{eq} : Equilibrium Weight

$$* w = (\lambda \Sigma)^{-1} \mu$$

Dynamic Portfolio Optimization

Stochastic Brownian Motion :

$$\frac{dS}{S} = \mu dt + \sigma dX$$

$$\begin{aligned} dS_1 &= \mu_1 S_1 dt + \sigma_1 S_1 dX_1 \\ &\vdots \\ dS_n &= \mu_n S_n dt + \sigma_n S_n dX_n \end{aligned}$$

Investor's Wealth :

$$dV(t) = V(t) h_0(t) r_{money}(t) + \sum_{i=1}^N V(t) h_i(t) r_{risky}(t)$$

$$\begin{array}{c} | \\ t = 0 \quad \text{-----} \quad t = T \end{array}$$

Initial Wealth : $V(0) = V$

Weight : $h_0(t) + \sum_i h_i(t) = 1$

Stopping Time : $\tau = \inf \{t > 0 \mid V_t = 0\} \wedge T \longrightarrow \text{Utility} = 0$

Objective Function :

$$\int_{t=0}^T \phi(t, V_T) dt = \int_{t=0}^T \mathbb{E} [U(V(t))] dt$$

Utility Function : $U(.)$

Value Function : $\phi(t, V_T)$

Quantum Computation Portfolio Optimization

Quantum State :

$$|\Psi(\theta)\rangle = \alpha|0\rangle + \beta|1\rangle$$

Objective Function :

$$\min_{\theta} \langle \Psi(\theta) | H | \Psi(\theta) \rangle$$

$$H : \text{Hamiltonian} \longrightarrow H = w^T \Sigma w - w^T \mu$$

N = 4 Assets

$$|\Psi(\theta)\rangle = |0000\rangle$$

Quantum Circuit

Quantum
Measurement

Classical
Optimization

Upgrade θ to have energy converge to ground state